Quiz 1

Student Name: ____
Student ID: ____

A factory produces widgets from **three machines**: Machine 1 (M1), Machine 2 (M2), and Machine 3 (M3). Some widgets are defective.

Define events:

- M_i: widget comes from Machine i
- D: widget is defective

The **conditional probabilities** are:

Machine	Conditional Probabilities
M1	$P(M_1 \mid D) = 0.2$, $P(D \mid M_1) = 0.01$
M2	$P(M_2 \mid D) = 0.3$, $P(D \mid M_2) = 0.03$
M3	$P(M_3 \mid D) = 0.5$, $P(D \mid M_3) = 0.05$

Instructions

- Write all **relevant probability formulas** (law of total probability, Bayes' theorem, independence definition) **before substituting numbers**.
- Use clear mathematical notation and show all steps.

Questions

1. **(4 pts)** Compute the **probability that a randomly selected widget is defective**, $P(D)$.
2. **(4 pts)** Using Bayes' theorem, determine the **proportion of widgets produced** by M1, M2, and M3 $(P(M_1), P(M_2), P(M_3))$.
3. **(4 pts)** Compute the probability that a widget comes from each machine **and is defective**, i.e., $P(M_1 \cap D)$, $P(M_2 \cap D)$, $P(M_3 \cap D)$.
4. **(4 pts)** For the events $M_1 \cap D$, $M_2 \cap D$, $M_3 \cap D$:
   - Are they **mutually exclusive**?
   - Are any of them **independent**? Justify your answer.
5. **(4 pts)** If **two widgets are randomly selected with replacement**, what is the probability that **at least one is defective**?